BROWSER-BASED HANDS-ON

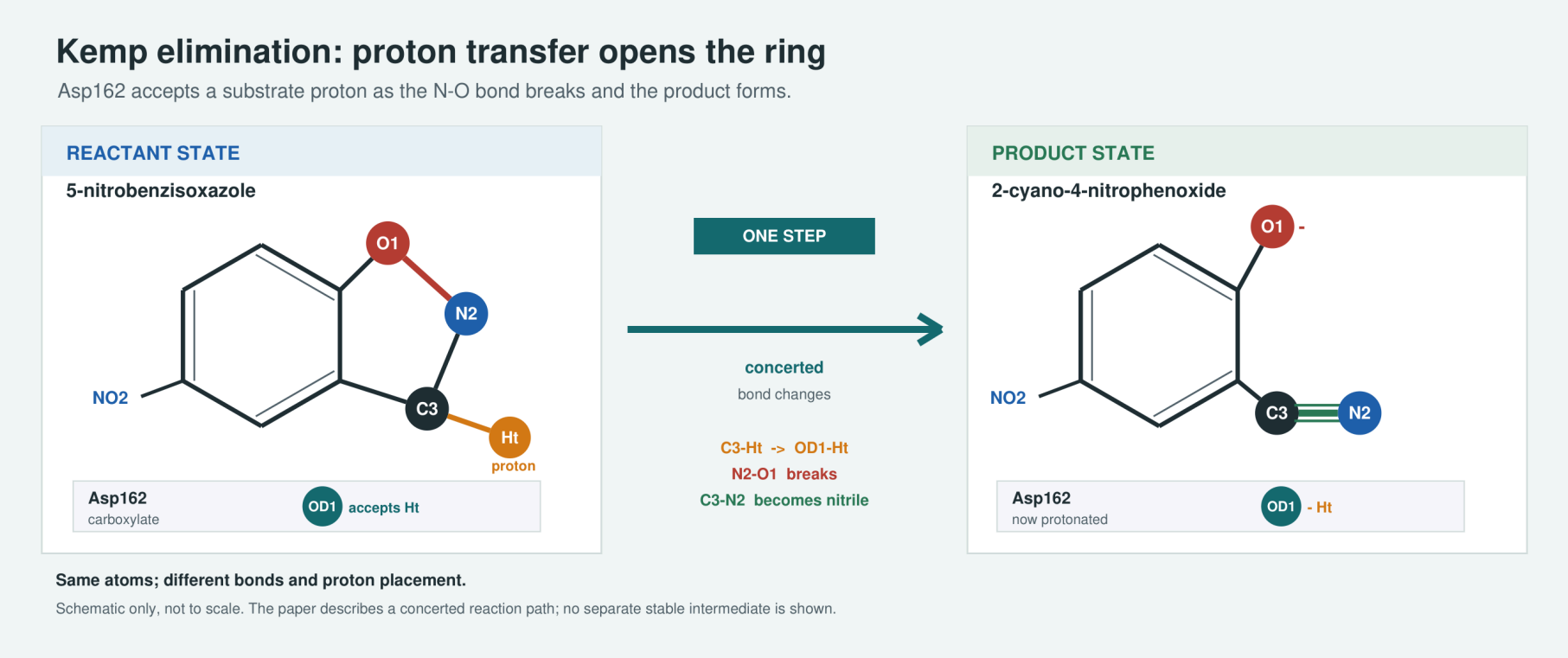
Try EVB with the Kemp reaction

Let's look at the Kemp reaction. We'll compare reactant and product, then change two EVB settings and see how the profile responds.

Workshop website: nd7996.github.io/Kemp-EVB-Workshop

What changes?

The substrate is the molecule the enzyme acts on. Asp162, part of the enzyme, accepts a proton (a hydrogen nucleus) from it. As the C–H bond breaks, the small ring opens and the product forms.



Two descriptions of the same atoms

REACTANT STATE

Before

The substrate holds the proton and its ring is closed. Asp162 is ready to accept it.

PRODUCT STATE

After

Asp162 holds the proton. The substrate ring is open.

EVB uses both descriptions to model the bond changes.

ATOM LABELS

C3 is the substrate carbon that loses Ht; Ht is the transferred hydrogen; O1 and N2 are the ring oxygen and nitrogen. Asp162 is enzyme aspartate 162; OD1 is its accepting side-chain oxygen.

WHAT THE METHODS GIVE

Q6 reports activation-barrier estimates for enzyme variants. A converged OpenMM run gives a free-energy profile, barrier, and reaction free energy. The saved profile is unvalidated.

What do the settings mean?

TWO STATES

Reactant and product

Each has its own bond and charge description for the same atoms.

A

Energy offset

Moves the product-state energy up or down relative to the reactant.

H12

Coupling

Controls how strongly the reactant and product descriptions mix.

Neither α nor H12 is the reaction barrier. Both affect the profile, so real calculations need evidence to choose them.

4 | READ THE COMPARISON CAREFULLY

These plots show different things

THE PAPER | FIGURE 3e

Q6 activation barriers

Compares experimental and Q6 activation-barrier estimates for Des27 and Des27.7, including the in and out substrate conformations. This is a barrier comparison, not a free-energy profile curve.

Open complete Figure 3 from paper

THIS WORKSHOP | OPENMM

Short-run free-energy profile

Shows the saved OpenMM adaptation: 21 windows, 50 records per window, and 18 analysis bins. Its 9.57 kcal/mol peak estimate is unvalidated; convergence has not been established.

Open OpenMM profile viewer

Q6 used a surface-constrained solvent sphere and local reaction field. The OpenMM adaptation differs in boundary treatment, electrostatics, Morse energy reference, and sampling. That explains why the results are not a direct engine comparison.

Prepare, relax, then sample

PREPARE

Build R and P

Use the same atoms and atom order. Assign each state's bonds, charges, force-field terms, and EVB settings. Check mapping, total charge, proton position, solvent, and ions.

RELAX

Settle both endpoints

Minimize and equilibrate R and P separately. Check geometry and stability. This prepares the endpoints; it does not calculate the reaction barrier.

FEP / PROFILE

Sample lambda windows

Sample the path between R and P. Check window overlap and convergence, compare replicas, then analyze the profile, barrier, and reaction free energy.

This workshop includes saved Kemp data for learning the analysis, not the full live scripts for these three simulation stages.

6 | YOUR TURN

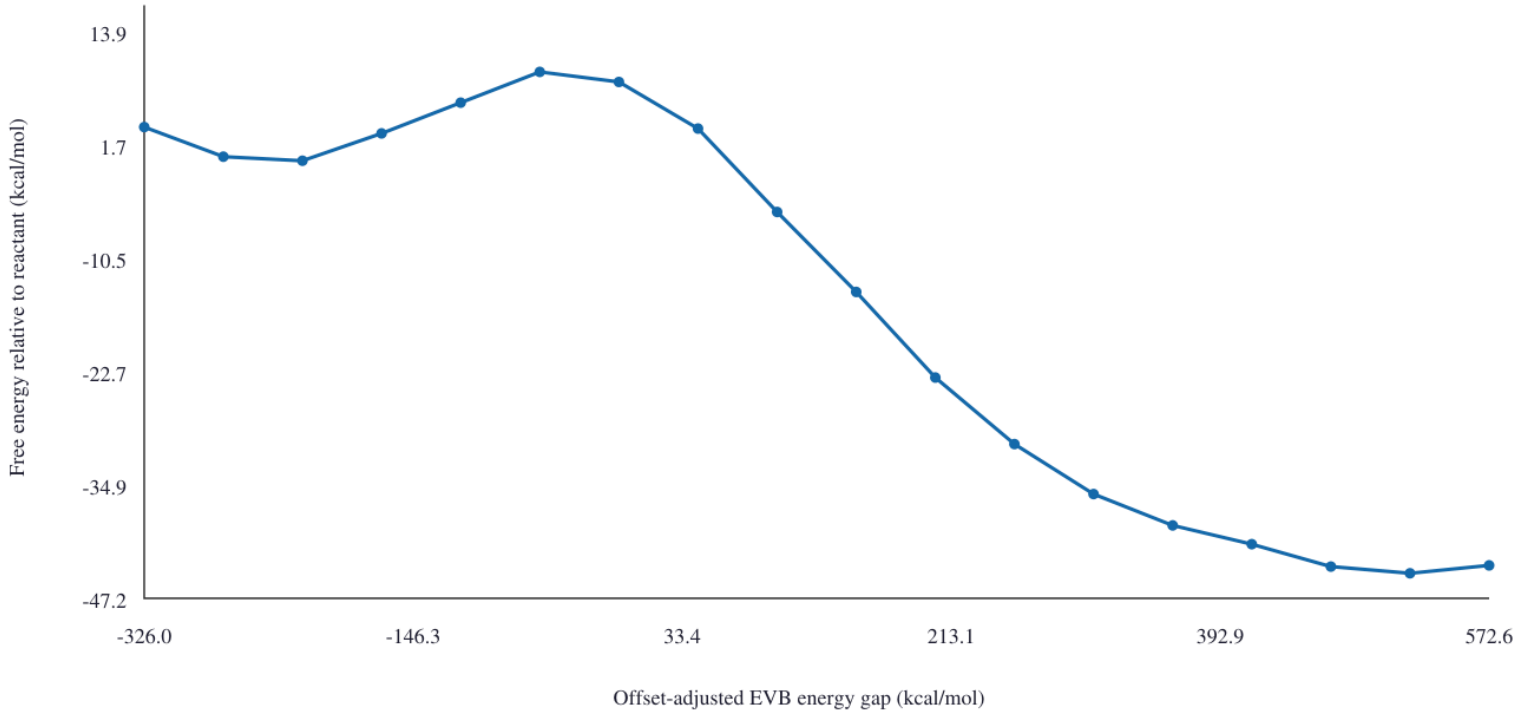
Try changing the settings

Open the small EVB example. Move one control at a time. Watch how the curve changes when you change the energy offset, then when you change the coupling.

Open the EVB example

Take about 5 minutes. The small example teaches the controls; it is not the full enzyme calculation.

The saved reaction profile



Saved short-run profile; not validated.

One barrier does not choose one model

Try the parameter comparison. Different offset/coupling pairs can give similar peak heights while giving different product-versus-reactant energies.

Open the settings comparison

These are teaching examples made from saved data. Matching a target is not a parameter fit or a validated prediction.

What should you take away?

- EVB connects reactant and product bonding descriptions.
- The offset and coupling shape the profile in different ways.
- Reference data and adequate sampling are needed before interpreting a profile physically.

Listov et al., Nature (2025). DOI: 10.1038/s41586-025-09136-2